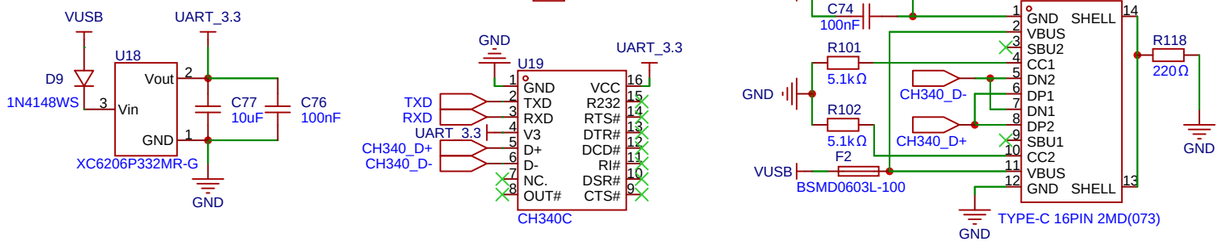
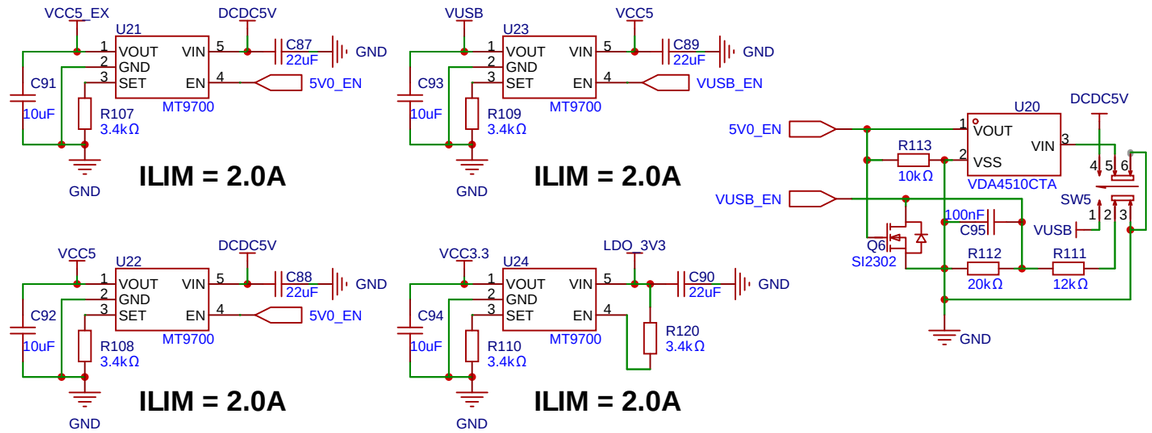
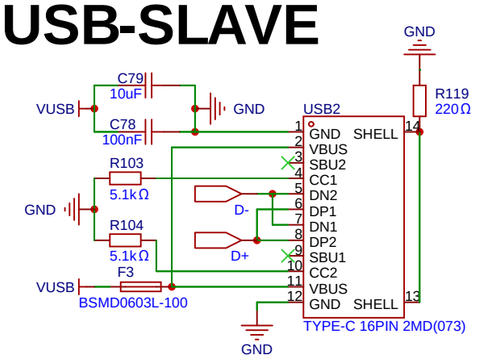
USART



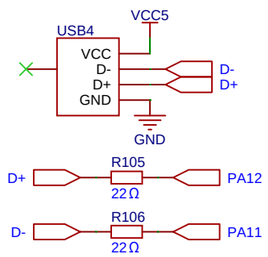
POWER PROTECTION

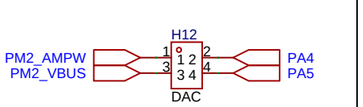


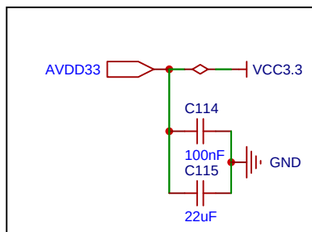
USB-SLAVE



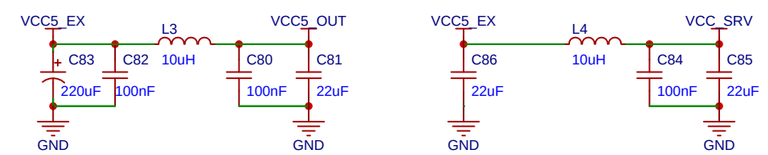
USB_HOST



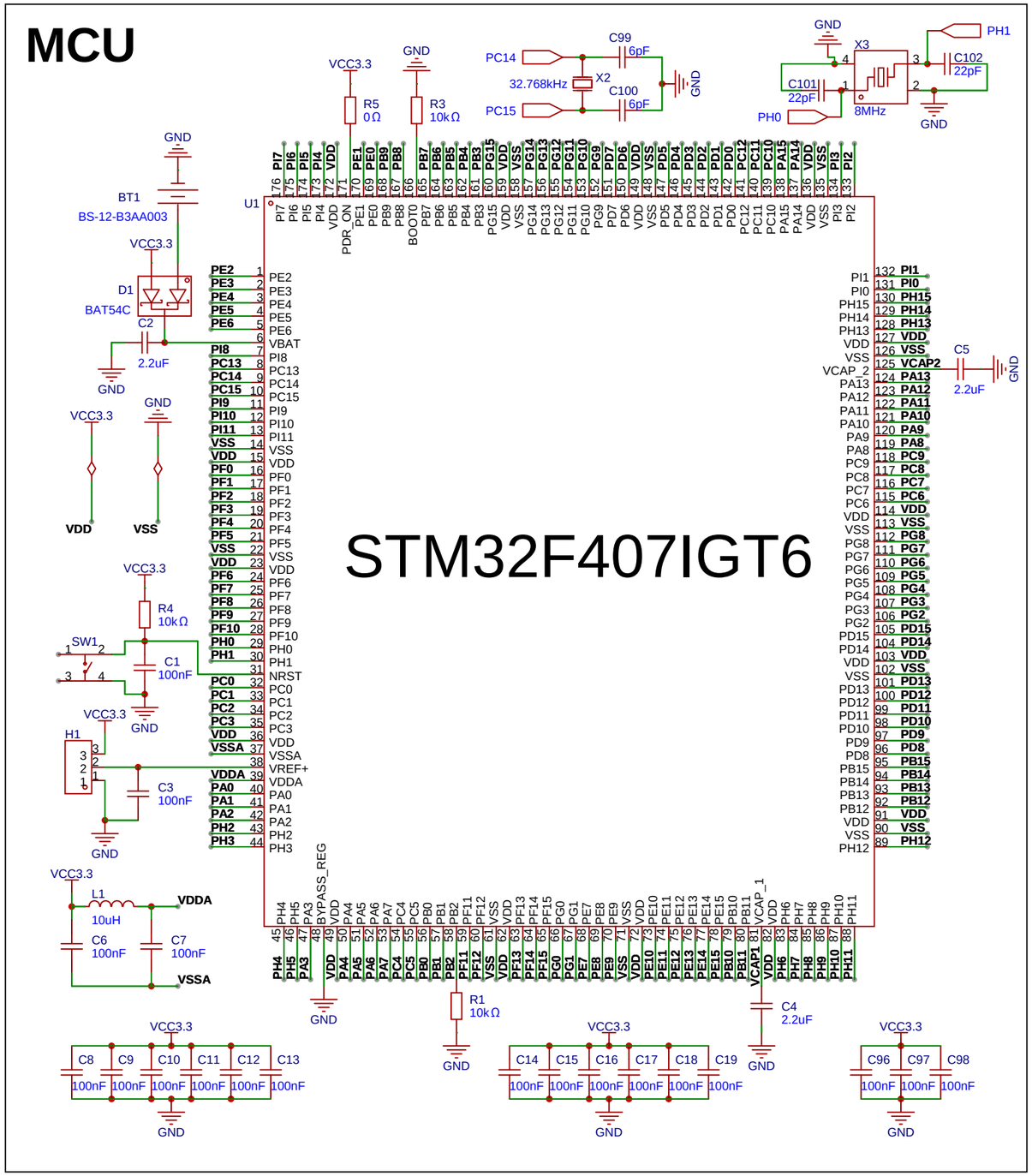




5V_EX

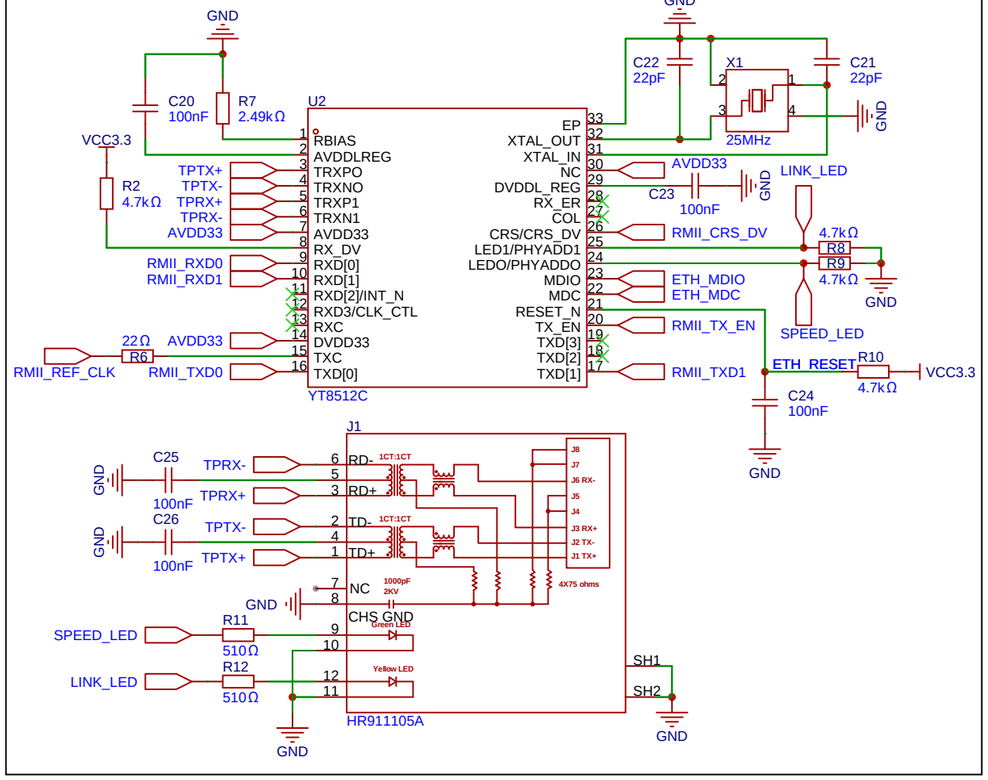


MCU

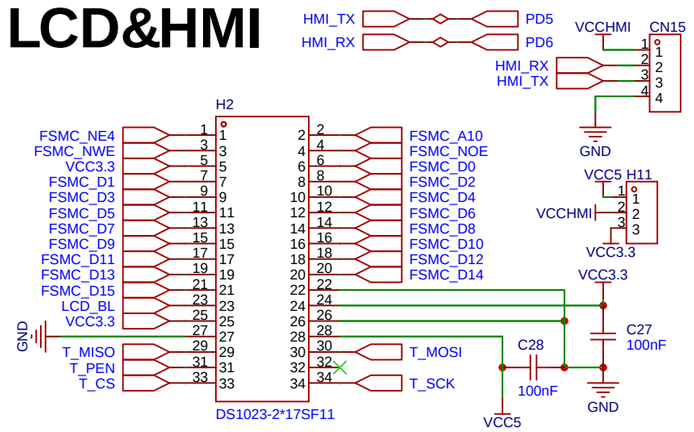


原理图	Schematic1_1	更新日期	2024-12-16
		创建日期	2024-11-22
图页	P1	物料编码	
绘制			
审阅			
	DRG-电机控制板-STM32F4-V1		
		版本	V1.0
		尺寸	A4
		页 1 共 4	
嘉立创EDA		嘉立创EDA	

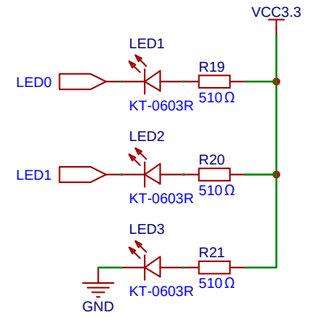
EARTHNET



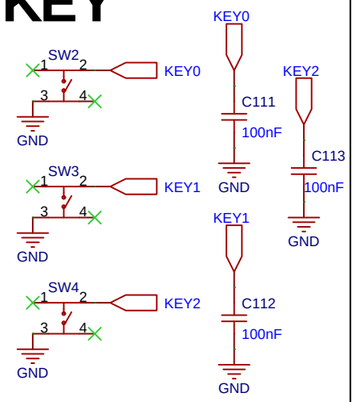
LCD&HMI



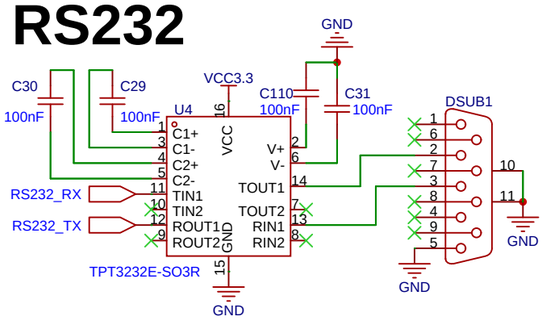
LED



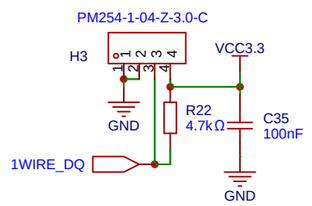
KEY



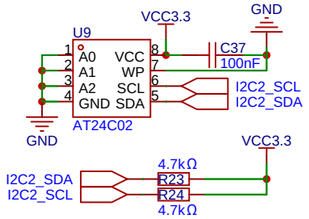
RS232



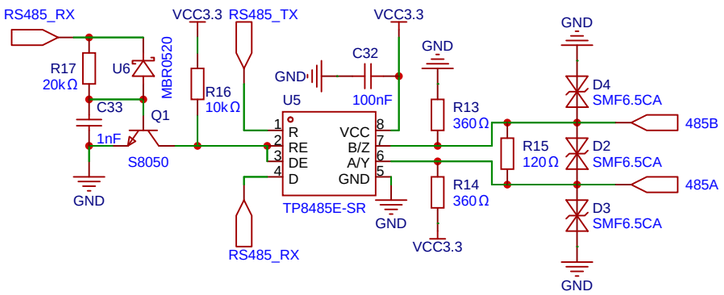
DHT11/DS18B20



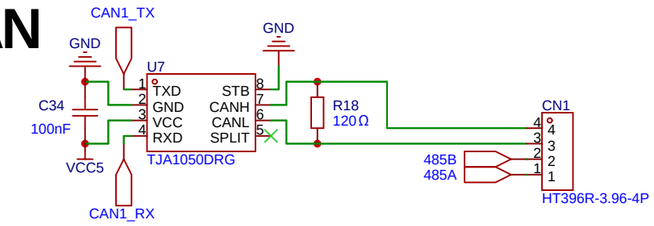
EEPROM



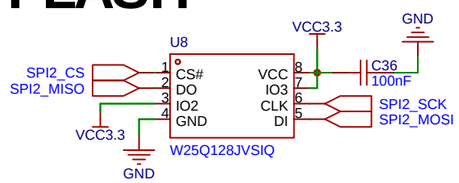
RS485



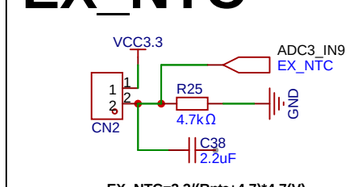
CAN



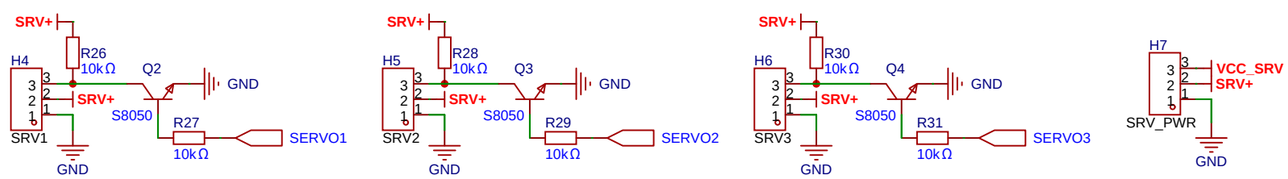
FLASH



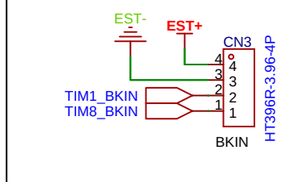
EX_NTC



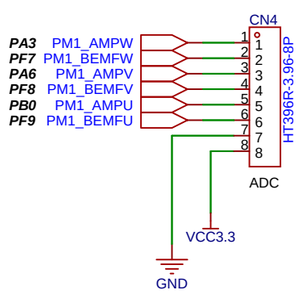
SERVO



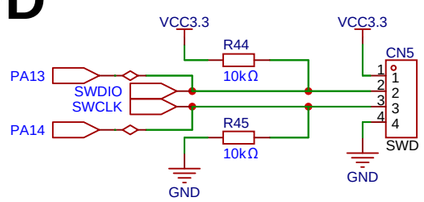
TIM_BKIN



ADC

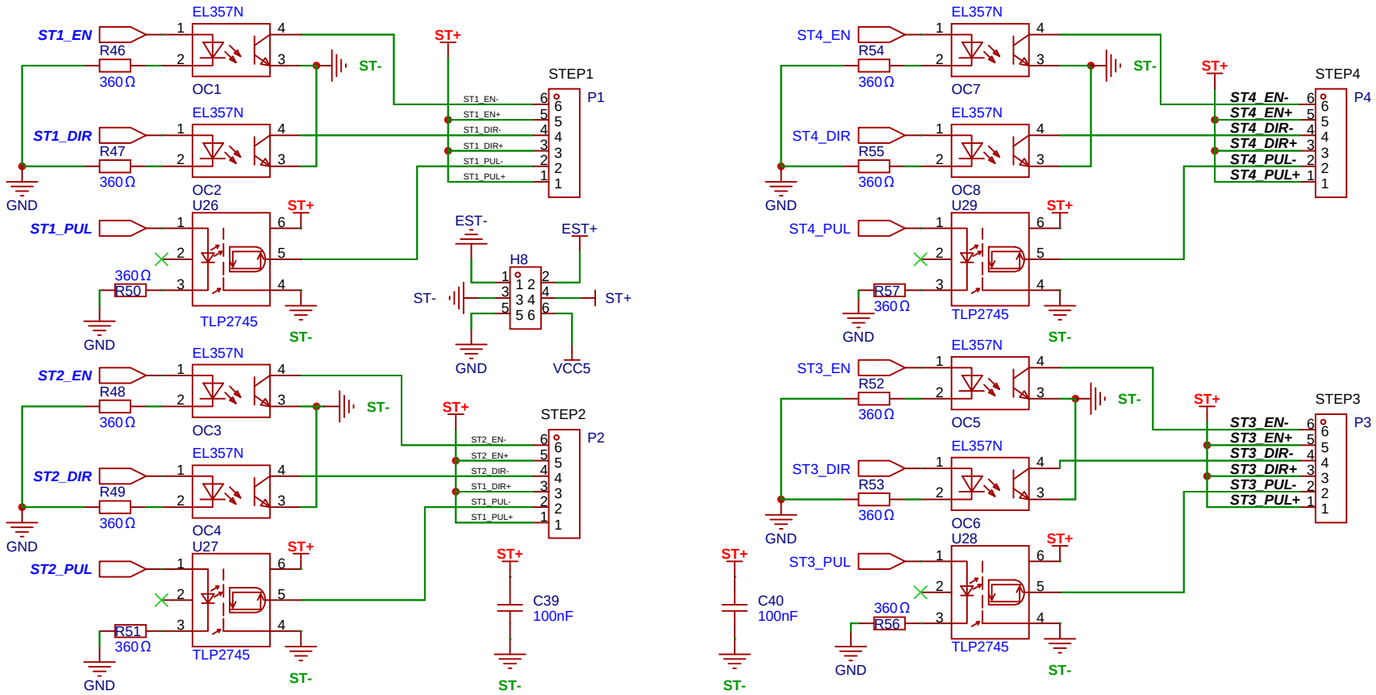


SWD

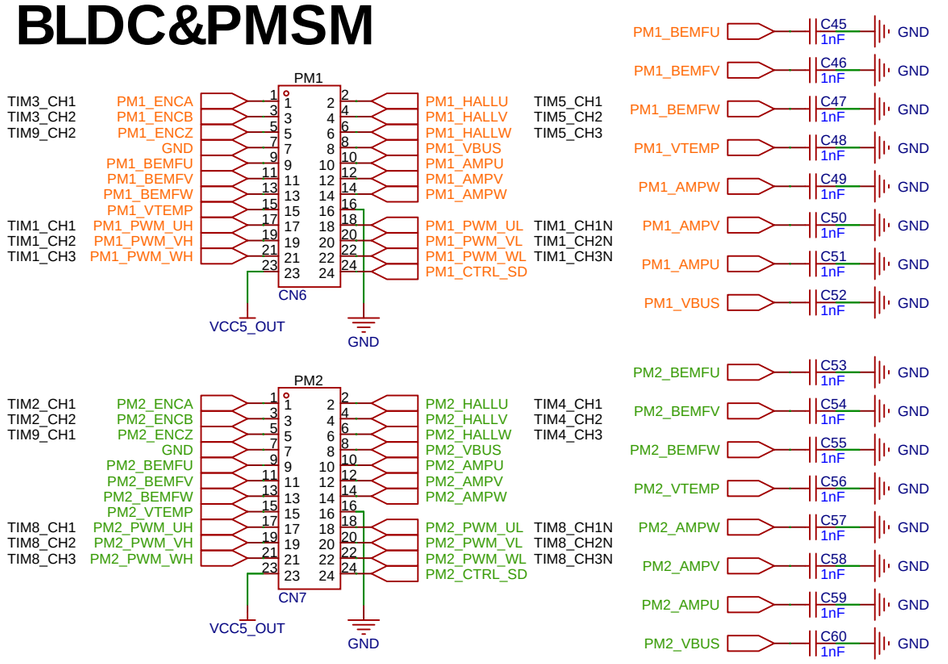


原理图	Schematic1_1	更新日期	2024-12-16
图页	P2	创建日期	2024-11-22
绘制		物料编码	
审阅		DRG-电机控制板-STM32F4-V1	
		版本	V1.0
		尺寸	A4
		页 2 共 4	
嘉立创EDA		嘉立创EDA	

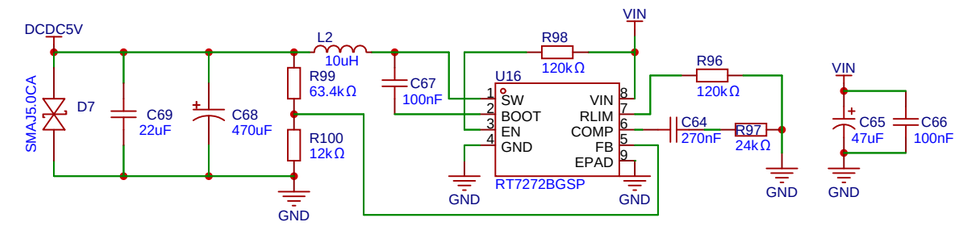
STEPMOTOR



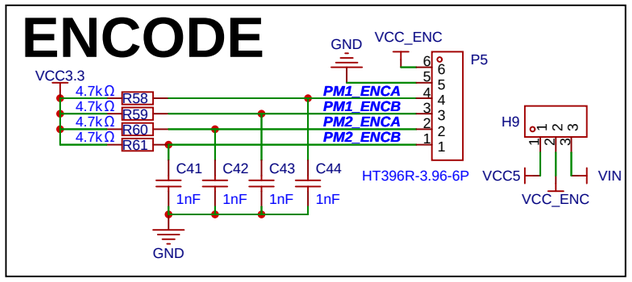
BLDC&PMSM



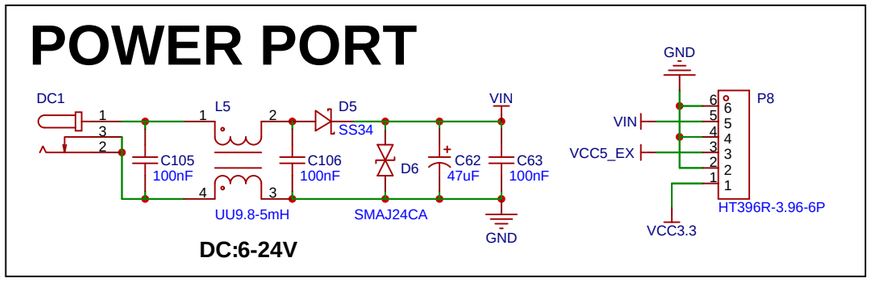
DC-DC 5V



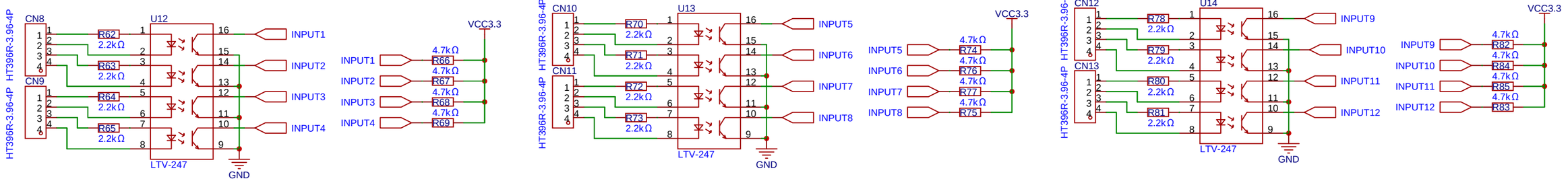
ENCODE



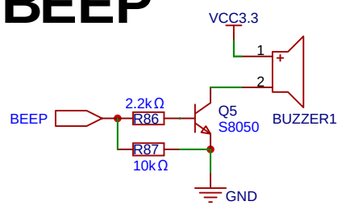
POWER PORT



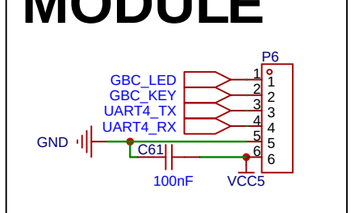
INPUT



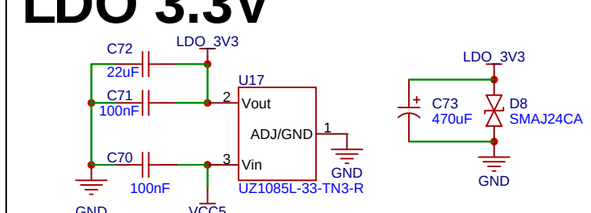
BEEP



MODULE



LDO 3.3V



原理图	Schematic1_1	更新日期	2025-01-04
		创建日期	2024-11-22
图页	P3	物料编码	
绘制			
审阅			
	DRG-电机控制板-STM32F4-V1		
	版本	尺寸	页 3 共 4
嘉立创EDA	V1.0	A4	嘉立创EDA

GPIOA

PA0	ADC123_IN0	
	PM1_VTEMP	
PA1	RMII_REF_CLK	
PA2	ETH_MDIO	
PA3	ADC123_IN3	
	PM1_AMPW	
PA4	ADC12_IN4	
PA5	ADC12_IN5	
PA6	ADC12_IN6	
	PM1_AMPV	
PA7	RMII_CRS_DV	
PA8	TIM1_CH1	
	PM1_PWM_UH	
PA9	TIM1_CH2	
	PM1_PWM_VH	
PA10	TIM1_CH3	
	PM1_PWM_WH	
PA11	USBD-	USB
PA12	USBD+	
PA13	SWDIO	SWD
PA14	SWCLK	
PA15	TIM2_CH1	
	PM2_ENCA	

GPIOB

PB0	ADC12_IN8
	PM1_AMPU
PB1	ADC12_IN9
	PM1_VBUS
PB2	BOOT1
	ST3_DIR
PB3	TIM2_CH2
	PM2_ENCB
PB4	OUTPUT1
PB5	OUTPUT2
PB6	USART1_TX
PB7	USART1_RX
PB8	TIM4_CH3
	PM2_HALLW
PB9	CAN1_TX
PB10	USART3_TX
	RS485_RX
PB11	USART3_RX
	RS485_TX
PB12	TIM1_BKIN
	TIM1_BKIN
PB13	TIM1_CH1N
	PM1_PWM_UL
PB14	TIM1_CH2N
	PM1_PWM_VL
PB15	TIM1_CH3N
	PM1_PWM_WL

GPIOC

PC0	ADC123_IN10
	PM2_VTEMP
PC1	ETH_MDC
PC2	ADC123_IN12
	PM2_AMPU
PC3	ADC123_IN13
	PM2_AMPV
PC4	RMII_RXD0
PC5	RMII_RXD1
PC6	TIM3_CH1
	PM1_ENCA
PC7	TIM3_CH2
	PM1_ENCB
PC8	INPUT8
PC9	TIM8_CH4
	ST4_PUL
PC10	UART4_TX
PC11	UART4_RX
PC12	UART5_TX
	RS232_RX
PC13	OUTPUT4
PC14	OSC32_IN
PC15	OSC32_OUT

GPIOD

PD0	FSMC_D2
PD1	FSMC_D3
PD2	UART5_RX
	RS232_TX
PD3	INPUT9
PD4	FSMC_NOE
PD5	FSMC_NWE
	HMI_TX
PD6	HMI_RX
PD7	INPUT10
PD8	FSMC_D13
PD9	FSMC_D14
PD10	FSMC_D15
PD11	T_MISO
PD12	TIM4_CH1
	PM2_HALLU
PD13	TIM4_CH2
	PM2_HALLV
PD14	FSMC_D0
PD15	FSMC_D1

GPIOH

PH0	OSC_IN
PH1	OSC_OUT
PH2	ST4_DIR
PH3	ST4_EN
PH4	I2C2_SCL
PH5	I2C2_SDA
PH6	TIM12_CH1
	T_SCK
PH7	T_PEN
PH8	T_MOSI
PH9	TIM12_CH2
	LCD_BL
PH10	TIM5_CH1
	PM1_HALLU
PH11	TIM5_CH2
	PM1_HALLV
PH12	TIM5_CH3
	PM1_HALLW
PH13	TIM8_CH1N
	PM2_PWM_UL
PH14	TIM8_CH2N
	PM2_PWM_VL
PH15	TIM8_CH3N
	PM2_PWM_WL

GPIOF

PF0	BEEP
PF1	OUTPUT3
PF2	PM2_CTRL_SD
PF3	ADC3_IN9
	EX_NTC
PF4	ADC3_IN14
	PM2_BEMFW
PF5	ADC3_IN15
	PM2_BEMFV
PF6	ADC3_IN4
	PM2_BEMFU
PF7	ADC3_IN5
	PM1_BEMFW
PF8	ADC3_IN6
	PM1_BEMFV
PF9	ADC3_IN7
	PM1_BEMFU
PF10	ADC3_IN8
	PM1_CTRL_SD
PF11	ST3_EN
PF12	ST2_DIR
PF13	ST2_EN
PF14	ST1_DIR
PF15	ST1_EN

GPIOE

PE0	LED0
PE1	LED1
PE2	KEY0
PE3	KEY1
PE4	KEY2
PE5	TIM9_CH1
	PM2_ENCZ
PE6	TIM9_CH2
	PM1_ENCZ
PE7	FSMC_D4
PE8	FSMC_D5
PE9	FSMC_D6
PE10	FSMC_D7
PE11	FSMC_D8
PE12	FSMC_D9
PE13	FSMC_D10
PE14	FSMC_D11
PE15	FSMC_D12

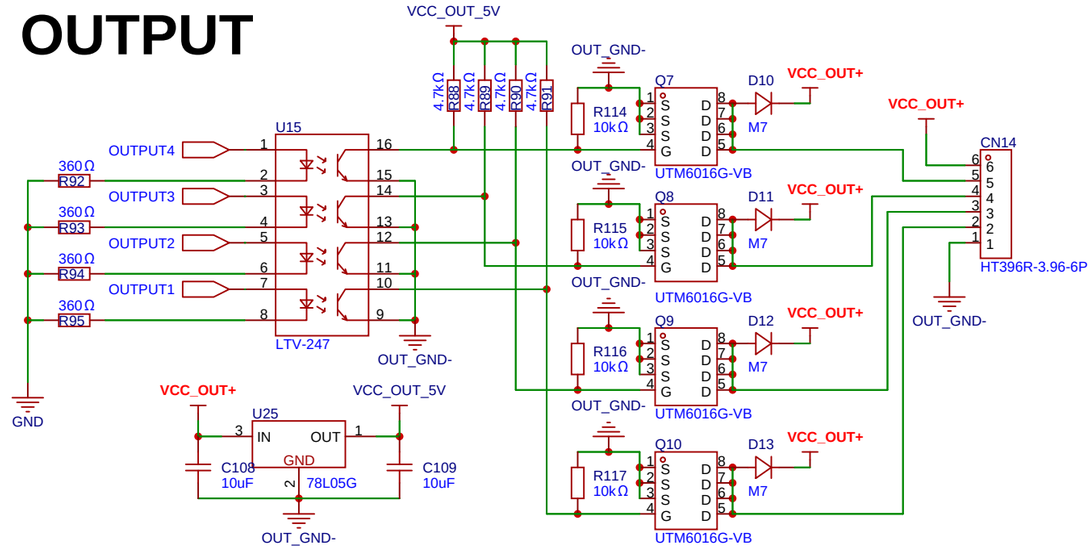
GPIOI

PI0	SPI2_CS
PI1	SPI2_SCK
PI2	SPI2_MISO
PI3	SPI2_MOSI
PI4	TIM8_BKIN
	TIM8_BKIN
PI5	TIM8_CH1
	PM2_PWM_UH
	ST1_PUL
	SERVO1
PI6	TIM8_CH2
	PM2_PWM_VH
	ST2_PUL
	SERVO2
PI7	TIM8_CH3
	PM2_PWM_WH
	ST3_PUL
	SERVO3
PI8	ETH_RESET
PI9	CAN1_RX
PI10	GBC_KEY
PI11	GBC_LED

GPIOG

PG0	FSMC_A10
PG1	T_CS
PG2	INPUT1
PG3	INPUT2
PG4	INPUT3
PG5	INPUT4
PG6	INPUT5
PG7	INPUT6
PG8	INPUT7
PG9	INPUT11
PG10	INPUT12
PG11	RMII_TX_EN
PG12	FSMC_NE4
PG13	RMII_TXD0
PG14	RMII_TXD1
PG15	1WIRE_DQ

OUTPUT



原理图	Schematic1_1	更新日期	2024-12-16
		创建日期	2024-11-22
图页	P4	物料编码	
绘制		DRG-电机控制板-STM32F4-V1	
审阅			
		版本	尺寸	页 4 共 4
嘉立创EDA		V1.0	A4	嘉立创EDA